ALKEM | AI MASTERCLASS

GenAI for Patient-Centric Healthcare

Participant Workbook

A practical workbook for doctors, pharmacists, medical affairs, pharmacovigilance, research, quality, and healthcare leadership.

60 min Concise theory and medical demonstrations	**90 min** Six case-based labs with verification
4 pillars Patient • Clinical • Pharma/Research • Governance	**Safety** Synthetic data • bounded task • qualified human decision

Important: Every case, image, name, and value in this kit is synthetic. The materials are for education, not diagnosis, prescribing, treatment, regulatory submission, or autonomous decision-making.

1. Four-pillar map and learning outcomes

Pillar	Purpose	Medical examples
01 Patient engagement	Make information clearer and actionable.	Prescription explanation; discharge teach-back; multilingual counselling
02 Clinical workflow	Structure information while clinicians retain decisions.	Image observation boundary; red flags; documentation
03 Pharma and research	Improve traceability and preparation.	PV intake; evidence search; medical information
04 Responsible AI	Match controls to risk and implementation.	Risk tier; data boundary; monitoring; stop rule

By the end, I can...

☐ frame a bounded healthcare task and name prohibited actions

☐ select a tool by workflow fit, data boundary, evidence, and reviewer

☐ use synthetic prescriptions and medical images safely in demonstrations

☐ verify source fidelity, citations, missingness, uncertainty, and escalation

☐ design a measurable, reversible 30-day pilot

My safety contract
Use synthetic or approved data. Do not delegate diagnosis, prescribing, safety, regulatory, publication, or governance decisions. Record accept, edit, reject, or escalate.

2. Workflow-first prompt canvas

Element	Question	My draft
Role	Who is the model assisting?	
Context	Audience, setting, approved data, and source?	
Task	One bounded action?	
Constraints	What must not be invented or decided?	
Output	What exact structure is useful?	
Verify	What source checks and escalation are required?	

Prompt quality check

A confident tone is not evidence. A safer prompt keeps source facts, assumptions, interpretation, missing data, and human decisions separate.

Lab 1 — Prescription extraction and explanation

Time	Action
3 min	Orient to the source and boundary
6 min	Run the bounded prompt
4 min	Verify against source
2 min	Accept, edit, reject, or escalate

Dataset: prescription_cases.csv

Goal: Exact extraction • [UNREADABLE] instead of guessing • plain-language explanation

Verification

☐ Every medicine/product and strength matches the source

☐ Frequency, duration, and warnings match the source

☐ No indication, interaction, substitution, or new advice was added

☐ A clinician or pharmacist owns the final decision

Human decision record

Decision	Evidence checked / edits / escalation
☐ Accept ☐ Edit ☐ Reject ☐ Escalate	

Lab 2 — Medical-image observation boundary

Time	Action
3 min	Orient to the source and boundary
6 min	Run the bounded prompt
4 min	Verify against source
2 min	Accept, edit, reject, or escalate

Dataset: imaging_cases.csv

Goal: Image adequacy • neutral observations • uncertainty • escalation—not diagnosis

Verification

☐ Adequacy and limitations assessed first

☐ No diagnosis, emergency clearance, or treatment recommendation

☐ Cannot-assess items and uncertainty are explicit

☐ Qualified clinical interpretation is explicit

Human decision record

Decision	Evidence checked / edits / escalation
☐ Accept ☐ Edit ☐ Reject ☐ Escalate	

Lab 3 — Discharge plan and multilingual teach-back

Time	Action
3 min	Orient to the source and boundary
6 min	Run the bounded prompt
4 min	Verify against source
2 min	Accept, edit, reject, or escalate

Dataset: discharge_summaries.csv

Goal: Preserve medicine, dose, date, warning, follow-up, and source meaning

Verification

☐ Clinical meaning is unchanged

☐ Urgent warning signs are prominent

☐ Teach-back questions reveal action comprehension

☐ Fluent clinical translation review is named

Human decision record

Decision	Evidence checked / edits / escalation
☐ Accept ☐ Edit ☐ Reject ☐ Escalate	

Lab 4 — Differential and red-flag checklist

Time	Action
3 min	Orient to the source and boundary
6 min	Run the bounded prompt
4 min	Verify against source
2 min	Accept, edit, reject, or escalate

Dataset: clinical_reasoning_cases.csv

Goal: Common • cannot-miss • evidence for/against • missing data • urgency

Verification

☐ No final diagnosis or treatment

☐ Common and cannot-miss items are separated

☐ Missing history/examination remains visible

☐ Cognitive-bias and escalation checks are present

Human decision record

Decision	Evidence checked / edits / escalation
☐ Accept ☐ Edit ☐ Reject ☐ Escalate	

Lab 5 — Pharmacovigilance intake

Time	Action
3 min	Orient to the source and boundary
6 min	Run the bounded prompt
4 min	Verify against source
2 min	Accept, edit, reject, or escalate

Dataset: adverse_event_reports.csv

Goal: Neutral chronology • seriousness fields • missing data • follow-up • PV route

Verification

☐ No causality, expectedness, coding, or reportability decision

☐ Seriousness rationale is traceable

☐ Minimum criteria and missing fields are visible

☐ Qualified PV/medical review is explicit

Human decision record

Decision	Evidence checked / edits / escalation
☐ Accept ☐ Edit ☐ Reject ☐ Escalate	

Lab 6 — Evidence search and citation verification

Time	Action
3 min	Orient to the source and boundary
6 min	Run the bounded prompt
4 min	Verify against source
2 min	Accept, edit, reject, or escalate

Dataset: research_questions.csv

Goal: PICO/PECO • reproducible search • screening log • open every source

Verification

☐ No fabricated paper or citation

☐ Search and screening decisions are reproducible

☐ Facts remain linked to opened sources

☐ Limitations and AI-use disclosure are present

Human decision record

Decision	Evidence checked / edits / escalation
☐ Accept ☐ Edit ☐ Reject ☐ Escalate	

9. Tool-choice and output comparison

Criterion	Tool A	Tool B	Tool C
Workflow/input fit			
Data/account boundary			
Source or evidence traceability			
Missingness and uncertainty			
Human correction required			
Access, region, and cost verified			
Decision: use / do not use / escalate			

Selection rule
Do not rank tools by fluency alone. Compare fidelity, traceability, safety boundaries, privacy fit, usability, and the amount of qualified correction required.

10. My 30-day responsible pilot

Canvas	My decision
Pain point and bounded task	
Accountable owner	
Allowed / prohibited data	
Users and qualified reviewer	
Baseline and success measure	
Monitoring and audit evidence	
Incident / escalation route	
Stop and rollback rule	

11. Sources and further reading

Accessed and verified for this masterclass on 28 July 2026. Product features, pricing, regional access, and policy status can change; recheck before delivery or implementation.

- **WHO: Ethics and governance of AI for health:** https://www.who.int/publications/i/item/9789240029200
- **NIST AI Risk Management Framework:** https://www.nist.gov/itl/ai-risk-management-framework
- **FDA: AI-enabled medical devices:** https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-enabled-medical-devices
- **FDA: Clinical decision support FAQ:** https://www.fda.gov/medical-devices/software-medical-device-samd/clinical-decision-support-software-frequently-asked-questions-faqs
- **ChatGPT image-input limitations:** https://help.openai.com/articles/8400551-chatgpt-image-inputs-faq
- **India Digital Personal Data Protection Act:** https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf
- **India Digital Personal Data Protection Rules:** https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa